TEMPO

From working Video RAG to measurable retrieval and object-centric identity analysis

Research development record | Part 2 of 3 | Stages 6–10

This installment continues the exact stage numbering established in Part 1. It documents the transition from a small working prototype to annotated long-video retrieval experiments, then to object detection, tracklet appearance matching, and an external identity benchmark. Each stage separates its intended goal, implementation, measured observations, limitations, and the reason for the next experiment.

Stage map

Stage	Research objective	Primary output
6	Obtain annotated longer videos	YouCook2 videos + event annotations
7	Measure event retrieval quantitatively	Recall@K and tolerant recall
8	Extract object tracks from videos	YOLO + ByteTrack track records
9	Link fragmented tracklets by appearance	CLIP crop embeddings and candidate links
10	Establish independent identity ground truth	BURST masks and identity-pair baseline

Evidence and interpretation

Numerical results below are drawn from the recorded experiments and available benchmark summaries. The five-stage division is an organizational reconstruction of the development history; it is not a claim that each stage was a formally frozen software release. Diagrams are conceptual unless identified as measured plots. We do not invent unavailable timings, GPU throughput, or missing confidence intervals.

Stage 6 | From short demonstrations to YouCook2

6.1 What we wanted to achieve

The first five stages demonstrated that a CLIP → FAISS → Qwen2.5-VL pipeline could retrieve frames and answer questions. The next question was whether it would remain useful on longer videos with multiple temporally localized actions. A research-quality evaluation needs videos, timestamps, and reference descriptions of what happens in each interval.

6.2 Why YouCook2?

YouCook2 provides cooking videos annotated with temporally segmented instructional steps. Each event has a start/end interval and a natural-language description. This is a natural testbed for text-to-video-moment retrieval: the text describes an action, and the system should return frames inside the annotated interval. The annotations are not perfect frame-by-frame labels; they are interval-level event references.

6.3 What we implemented

- Downloaded the YouCook2 train/validation annotation JSON; the relevant top-level field is database.
- An initial all-in-one download (~35 GB) did not provide usable MP4 files, so the workflow switched to per-video downloads.
- Fetched 19 of the first 20 targeted validation videos successfully; one source returned HTTP 404. Later, 20 annotated videos were available for batch processing.
- Mapped video identifiers to annotation records, filtered to locally available videos, and retained event start/end times for evaluation.

6.4 Observed results

The project identified 416 eligible validation videos in the annotation/download workflow. A representative video, --bv0V6ZjWI, had a recorded duration of 332.50 seconds at 30 FPS and 9,975 frames. At an approximately two-second sampling interval, 167 frames were extracted for retrieval. This established that the original frame-based architecture could handle a several-minute video without ingesting every source frame into CLIP.

6.5 Limitations and transition

Downloading video URLs is not fully reproducible when hosted media disappears; the 404 illustrates that problem. Cooking events may contain several visually similar states, and a sampled frame can fall just outside an annotation despite being semantically related. The next stage therefore formalized retrieval metrics rather than relying on a handful of plausible examples.

Stage 7 | Quantitative temporal retrieval evaluation

7.1 What we wanted to measure

For each annotated event, use its text description as a query, rank sampled video frames by CLIP similarity, and check whether any of the top K timestamps fall inside the annotated event interval. This isolates retrieval quality from VLM answer generation: a fluent answer cannot compensate for missing the relevant visual evidence.

7.2 Evaluation protocol

For event e with annotated interval [s_e, t_e], let r_1,...,r_K be the timestamps of the top-K retrieved frames. A hit occurs when at least one r_k lies in [s_e,t_e]. Recall@K is the fraction of events with a hit. A tolerant version expands the acceptable interval by ±2 seconds to accommodate sparse frame sampling and annotation-boundary uncertainty. A random-ranking baseline contextualizes performance.

Important: this is event-level retrieval recall, not classification recall, and not a measure of answer correctness. Each query is evaluated against the corresponding video, rather than a cross-video search corpus.

7.3 Single-video pilot

For --bv0V6ZjWI, the pilot used 167 extracted frames and 10 annotated events. Recorded Recall@1 was 0.30, Recall@3 was 0.60, and tolerant Recall@3 was 0.70. For the event “crush and chop the garlic” (ground-truth interval 8–23 s), the retrieved timestamps began [14, 20, 12, 22, 42] seconds: top-1, top-3, top-5, and top-10 all counted as hits for this event.

7.4 Batch experiment: 19 evaluated videos, 134 events

Metric	CLIP retrieval	Random baseline
Recall@1	0.3159	0.0795
Recall@3	0.4949	0.2149
Recall@5	0.5927	0.3250
Recall@10	0.6911	0.5239
Tolerant Recall@1 (±2 s)	0.3527	Not recorded
Tolerant Recall@3 (±2 s)	0.5182	Not recorded
Tolerant Recall@5 (±2 s)	0.6170	Not recorded
Tolerant Recall@10 (±2 s)	0.7352	Not recorded

The recorded batch summary describes 20 processed videos, but the aggregate recall figures above are reported for 19 evaluated videos and 134 annotated events. We preserve that distinction instead of treating the processing count as the evaluation denominator.

Stage 7 | Results and interpretation (continued)

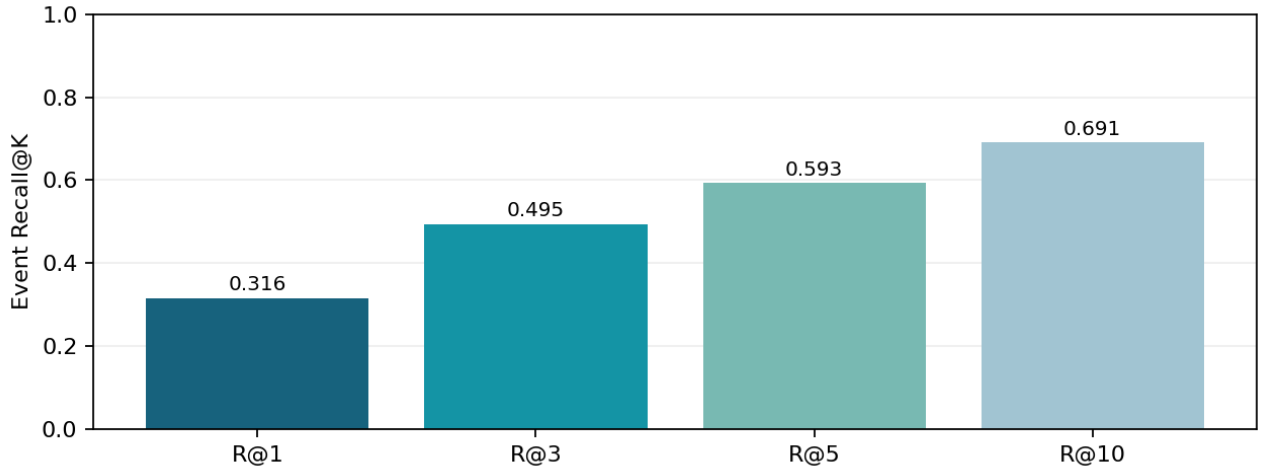


Figure 1. Measured YouCook2 event retrieval recall at increasing K (19 evaluated videos).

7.5 What the results tell us

- CLIP substantially outperformed random ranking at low K; Recall@1 was about four times the recorded random baseline.
- The gap narrows as K increases because retrieving more frames increases the chance of intersecting the annotated interval even for random retrieval.
- At K=10, roughly 31% of annotated events still had no strict hit, indicating that simple frame-level semantic similarity does not solve temporal localization.
- The ±2 s tolerance improved Recall@10 from 0.6911 to 0.7352; this shows some boundary sensitivity but does not explain all failures.

7.6 Why move beyond global frame embeddings?

A single CLIP embedding summarizes the full frame. It may emphasize the scene, background, or dominant object rather than the specific entity involved in a queried action. It also lacks an explicit persistent object identity. To support questions such as “what happened to the same object later?”, the system needs object-centric representations and temporal correspondence. This motivated object tracking.

Stage 8 | Object detection and ByteTrack tracking

8.1 What we wanted to achieve

We wanted to replace isolated frame descriptions with persistent object observations: bounding boxes, category predictions, track IDs, confidence scores, and timestamps. The first baseline used an off-the-shelf detector/tracker rather than training a bespoke model, allowing us to inspect whether existing tracking already provides reliable long-term entity memory.

8.2 Implementation

The object tracker used YOLO11n with ByteTrack through an interface equivalent to model.track(frame, persist=True, tracker="bytetrack.yaml", conf=0.25, verbose=False). It recorded frame index, timestamp, and detections with track_id, class_id, class_name, confidence, and bounding box. Output was saved in a tracks.json structure for downstream analysis.

8.3 Representative experiment

Measurement	Recorded result
Video	YouCook2 --bv0V6ZjWI
Processed frames	9,972
Total detections	26,822
Unique predicted track IDs	537
Mean observations per track	49.95
Median observations per track	7
Mean track duration	2.00 seconds
Median track duration	0.33 seconds
Tracks with fewer than 10 observations	306 (57%)

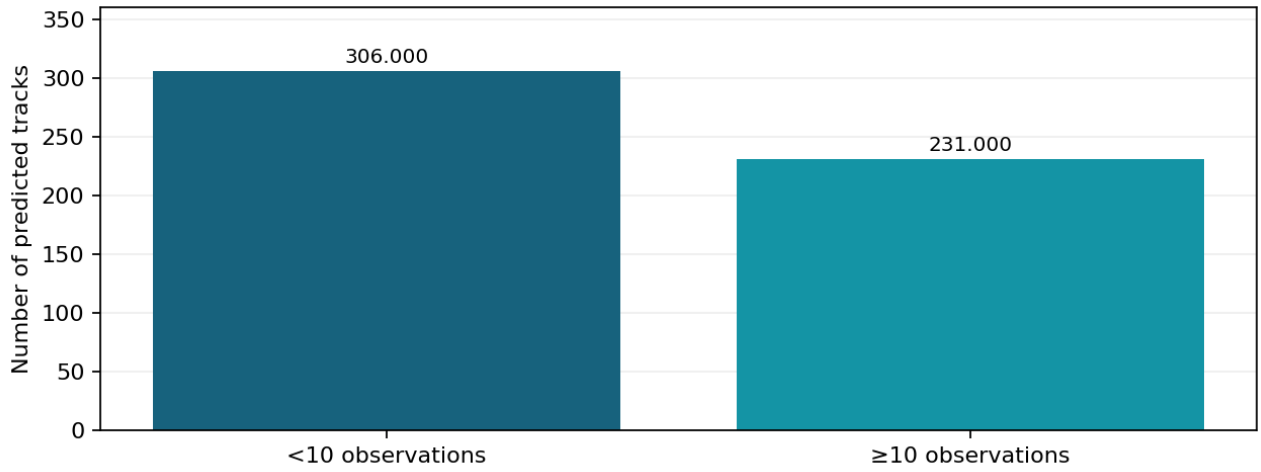


Figure 2. Measured track-length distribution grouped at 10 observations (537 tracks total).

8.4 What failed?

The detector frequently assigned unsuitable object categories in cooking scenes; one observed example was tomato slices classified as donuts. More importantly, the tracker generated many short-lived IDs. When a hand or object disappeared and returned, the same physical entity could receive a new track ID. These are related but distinct problems: category misclassification concerns object semantics, while ID fragmentation concerns temporal identity continuity.

8.5 Result and limitation

The pipeline produced machine-readable object tracks, but 537 track IDs are not 537 unique physical objects. The 57% short-track fraction indicated a substantial fragmentation problem. Without manually annotated identities, we could not yet quantify the exact re-identification error rate. We therefore explored appearance-based linking of fragmented tracklets.

Stage 9 | CLIP-based tracklet embeddings and linking

9.1 Research question

If ByteTrack assigns separate IDs to the same physical object after an occlusion or detection interruption, can an appearance embedding reconnect those fragments? We tested whether CLIP, already used for text/frame retrieval, could also represent the appearance of object crops.

9.2 Implementation

- Group detections by predicted track ID to construct a tracklet: a short sequence of bounding boxes for one provisional identity.
- Retain tracklets with more than 10 observations; sample five representative object crops per eligible tracklet.
- Encode each crop with CLIP; average the embeddings and L2-normalize to create a tracklet-level descriptor.
- Compute cosine similarity for every candidate pair, inspect top-ranked pairs, and record temporal overlap/gap for subsequent consistency checks.

9.3 Observed output

Measurement	Recorded result
Eligible tracklets	231
Candidate pairs	26,565
Example very high similarity	0.9971 (IDs 1521 and 1534)
Example long-gap high similarity	0.9928 (IDs 1030 and 2106; 105.07 s gap)
Another long-gap example	0.9926 (IDs 1580 and 2146; 60.33 s gap)

The user visually inspected the 20 highest-ranked candidate pairs and reported that they represented the same object, despite incorrect detector labels in several cases. This supported the plausibility of CLIP-based appearance linking, but the sample was small and strongly biased toward easy, high-similarity positives.

9.4 Why high cosine similarity is not enough

Two different objects of the same category may have nearly identical CLIP embeddings, particularly when they share color, shape, and scene context. Conversely, the same object may look different after a viewpoint change or occlusion. Temporal overlap provides a useful physical constraint: two distinct simultaneously visible objects cannot be merged into one identity. Thus appearance similarity alone is not a complete association rule.

9.5 Result and next step

Stage 9 generated promising candidate links but lacked an independent identity ground truth. Manually confirming only the top-ranked pairs cannot measure false merges among harder same-category negatives. The next stage introduced an annotated identity benchmark to test discrimination rigorously.

Stage 10 | BURST: an identity-grounded benchmark

10.1 What we wanted to achieve

We needed a benchmark in which the same physical object is labeled consistently across frames, including scenes with multiple objects of the same category. This allows quantitative evaluation of appearance-based re-identification rather than relying on a few manually reviewed matches. BURST supplies segmentation masks and temporal object IDs that can serve as oracle identity annotations.

10.2 Data acquisition and reproducibility

Downloaded BURST validation annotations from the official annotation archive and constructed a selected-sequence manifest. The initial eight Charades validation sequences contained 363 annotated frames, 1,147 object-mask observations, and 29 GT tracks. Source frames were selectively extracted from the TAO-hosted Charades ZIP using authenticated Hugging Face access and remote ZIP range requests; all 363 annotated frames were recovered. COCO-style RLE masks were decoded with pycocotools and verified through rendered overlays.

10.3 Identity baseline protocol

For each annotated observation, the script cropped the object using its ground-truth segmentation mask (with a gray background), encoded it with CLIP, and computed cosine similarities for pairs within the same video. A positive pair shares the same ground-truth track ID; a negative pair belongs to different GT tracks. The metrics were ROC-AUC and average precision (AP).

Important methodological distinction: GT masks, not detector boxes, define the crops. This isolates representation quality from detector errors. It is not yet an end-to-end tracker. GT identities label pairs for evaluation and are not input to the CLIP encoder.

10.4 Initial eight-Charades baseline

Metric	Result
Observations	1,147
Positive identity pairs	23,928
Negative identity pairs	61,471
Mean positive cosine	0.8915
Mean negative cosine	0.7576
ROC-AUC	0.8944
Average precision	0.8281

These initial results appeared strong, but a critical benchmark defect was discovered: the eight selected Charades sequences contained no same-category negative pairs. The model could often distinguish object classes rather than individual instances. AUC 0.8944 therefore overstated the relevance of the benchmark to same-category identity association.

Stage 10 | Hard-negative correction and results

10.5 Mining more discriminative sequences

The mining script scanned 988 BURST validation sequences and found 667 eligible sequences with co-visible, same-category objects. Twenty promising sequences were ranked for inspection. Five were selected and their annotated frames selectively extracted: LaSOT/volleyball-13 (person), LaSOT/person-17 (person), LaSOT/sheep-19 (sheep), HACS/Grooming_horse (horse), and HACS/Playing_badminton (person).

10.6 Hard identity benchmark

Metric	Measured result
Videos / observations	5 / 1,905
All pairs	354,842
Positive / negative pairs	35,636 / 319,206
All-pair ROC-AUC / AP	0.7553 / 0.3767
Same-category pairs	340,841
Same-category negatives	305,205
Same-category ROC-AUC / AP	0.7618 / 0.3873
Retrieval Recall@1 / Recall@5	0.7330 / 0.9223
Retrieval MRR	0.8127

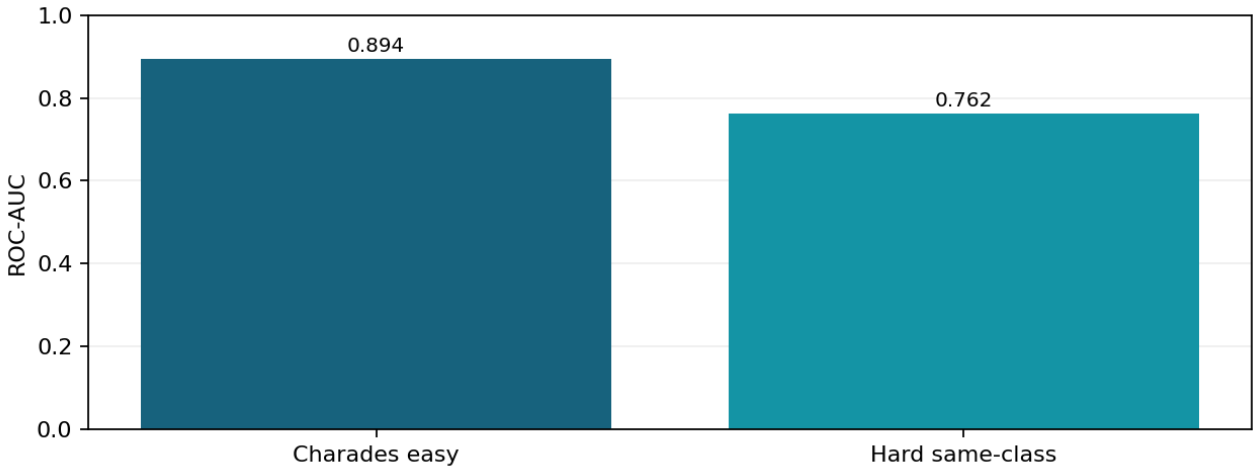


Figure 3. Recorded identity-pair AUC. The two evaluations use different sequence selections and negative-pair distributions; this is not a controlled same-test-set model comparison.

10.7 Temporal difficulty and sequence variation

Within the hard benchmark, identity-pair AUC decreased as the separation between annotated observations increased: 0.8083 for gaps 1–3, 0.7749 for 4–10, 0.7456 for 11–30, and 0.7018 for 31 or more annotated-frame positions. Same-category AUC was approximately 0.610 for Sheep-19 and 0.656 for Grooming_horse, versus 0.897 for Playing_badminton. This showed that appearance discriminability is highly scene-dependent.

10.8 What Stage 10 established

The initial benchmark was too easy; correcting the negative sampling exposed a much harder and more relevant re-identification problem. The main scientific result is not simply the lower AUC, but the discovery that instance-level identity association requires same-category negatives, temporal separation, and careful evaluation of ambiguity. This motivated the later failure analysis, uncertainty studies, oracle identity-memory experiments, and online association experiments covered in Part 3.

Synthesis | What Stages 6–10 changed

Stage	Result	What remained unsolved
6	Annotated long-video test material acquired	Unavailable videos and sampling gaps
7	CLIP Recall@10 = 0.6911 on YouCook2	Many events still missed; no object memory
8	537 predicted tracks; 57% very short	Fragmented IDs and detector mistakes
9	26,565 CLIP candidate links	No unbiased ground-truth validation
10	Hard same-category AUC = 0.7618	Appearance alone remains ambiguous

The conceptual shift

The project began with a retrieval question: “Which frame is relevant to this sentence?” It evolved toward an identity question: “Which observations across time refer to the same physical object?” The second question is essential for long-video temporal reasoning because an answer about what an object did earlier or later depends on maintaining its identity through changes in appearance, position, and visibility.

Interpretation guardrails

- YouCook2 event Recall@K, BURST pairwise ROC-AUC, and later online clustering F1 measure different tasks; they should not be plotted as though directly comparable.
- GT segmentation masks and GT category labels provide oracle perception in BURST; a deployment-ready method must eventually handle detection errors.
- Millions of correlated observation pairs are not independent experimental units. Video-level uncertainty and held-out sequence selection are necessary for publishable claims.
- Stages 6–10 demonstrate the need for identity memory but do not yet establish a novel association method.

Preview of Part 3

The final installment will trace how failure cases motivated uncertainty scoring, oracle identity memories, automatic causal association, Hungarian mutual exclusion, spatial-temporal matching, and confidence diagnostics. Because the research contains named sub-stages 11.7–12D, the last report should preserve those sub-stage names instead of collapsing them into misleadingly simple “Stage 11” and “Stage 12” summaries.

Provenance

This document was reconstructed from the recorded project discussions and experiment summaries available in this conversation. Numerical results are explicitly described as recorded observations. The conceptual interpretation and stage organization are explanatory synthesis, not independent replication.